Magneto-Coulombic Actuation System for Active Debris Removal

A Propellantless Approach to LEO Sustainability

Research Hackathon Submission

IIT Kanpur - Institute Research Symposium 2025

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The Space Debris Crisis

Low Earth Orbit Congestion:

- 200-2,000 km altitude faces critical debris accumulation
- Defunct satellites, rocket bodies, collision fragments
- **Kessler Syndrome** threat: cascading collisions

Mega-Constellation Impact:

- Starlink, OneWeb, Kuiper exponentially increasing orbital population
- Collision risk threatens long-term space sustainability

Mission Objective:

Remove non-cooperative 500mm cubesat debris while ensuring:

- Zero chaser damage
- No debris fragmentation
- Multiple debris removals per mission
- Fuel-efficient, sustainable operations

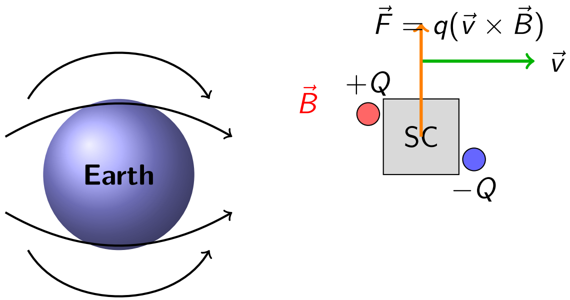
Limitations of Traditional Methods

Critical Challenges:

1. **Fuel Depletion**
   - Reaction thrusters rapidly consume propellant
   - Mission capacity: 1-2 debris per launch only
2. **Fragmentation Risk**
   - Thruster plumes ($>1000^\circ$C) damage debris during approach
   - Creates more fragments $\rightarrow$ worsens problem
3. **Contact Complexity**
   - Physical capture of tumbling debris requires precise synchronization
   - High fuel consumption for tumble-matching
4. **Economic Infeasibility**
   - Single-debris missions lack cost-effectiveness
   - Large-scale cleanup economically unviable

Core Innovation: Magneto-Coulombic Actuation

Paradigm Shift: Propellantless control using Earth's magnetic field



Principle: Electrically charged shells interact with Earth's geomagnetic field to generate **Lorentz forces** for attitude & orbital control

Physical Principles: The Lorentz Force

Governing Equation:

$$\boxed{\vec{F} = q(\vec{v} \times \vec{B})} \tag{1}$$

Parameters in LEO:

- q: Shell charge $= \pm 1\,\mu$C at ± 50 kV
- $\vec{v}$: Orbital velocity $\approx 7,670$ m/s
- $\vec{B}$: Earth's magnetic field $=$ 25,000-65,000 nT

Dual Control Capability:

Attitude Control:

$$\vec{\tau} = \sum_i \vec{r}_i \times [q_i(\vec{v} \times \vec{B})] \tag{2}$$

Shell pairs with opposite charges $(+Q, -Q)$ create torque for rotation

Orbital Maneuvering:

$$\vec{F}_{\text{net}} = Q_{\text{total}} \cdot \vec{v} \times \vec{B} \tag{3}$$

All shells charged identically create net force for gradual orbit changes

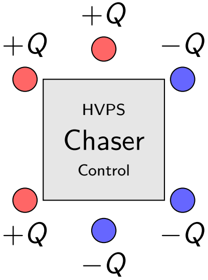
Spacecraft Configuration

Magneto-Coulombic System Design:

- **6 Coulomb shells** (0.5m $\times$ 0.5m plates) at spacecraft extremities
- **Charge capacity:** $\pm$50 kV, $\pm$1 μC per shell
- **High-voltage power supply (HVPS):** Cockcroft-Walton multipliers (28V DC $\rightarrow$ 50 kV)
- **Electron gun:** Field-emission cathode for debris charging (1-10 W, 1-100 μA)

System Budget:

- Total mass: 3.4 kg
- Power: 5-30 W
- TRL: 7-8 (heritage from ion thrusters)



Three-Option Debris Removal Strategy

Option 1: Origami Sail Deployment (High-Throughput)

- Carry 5 compact origami-folded drag sails (Miura-ori pattern)
- Each sail: 10m $\times$ 10m deployed from 0.5m cube (2 kg)
- Attach sail to debris $\rightarrow$ passive deorbit via drag (3-6 months)
- **Result: 5 debris per 35-day mission**

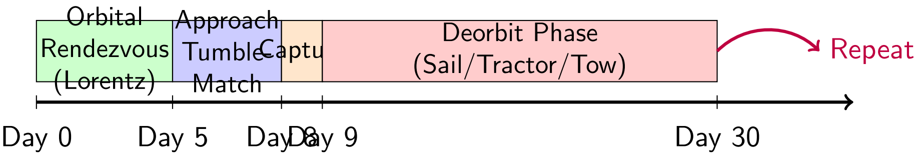
Option 2: Electrostatic Tractor (Touchless Removal)

- Charge chaser to $+50$ kV, debris to -30 kV via electron beam
- Maintain 5-10m station-keeping for 20 days
- At 5m: $F = \frac{kq_1 q_2}{r^2} \approx 3.6$ mN
- **Result: 2-3 debris with zero contact**

Option 3: Robotic Arm (Traditional Backup)

- 7-DOF arm with soft gripper/gecko-inspired adhesive
- Magneto-Coulombic provides precise pointing during capture
- **Result: High reliability for complex debris**

Mission Timeline: Hybrid Approach



Key Advantage: Magneto-Coulombic enables **minimal fuel consumption** during rendezvous and approach phases

Feasibility: Force & Torque Generation

Attitude Control Torque Calculation:

Example: Two shells at $r = 0.5$ m, $q = \pm 1 \times 10^{-6}$ C

$$F = q \cdot |v \times B| = (1 \times 10^{-6})(7670)(40 \times 10^{-6}) = 3.07 \times 10^{-7} \text{ N} \tag{4}$$

$$\tau = 2 \times r \times F = 2(0.5)(3.07 \times 10^{-7}) = 3.07 \times 10^{-7} \text{ N·m} \tag{5}$$

For spacecraft with $I = 10$ kg·m^2:

$$\alpha = \frac{\tau}{I} = 3.07 \times 10^{-8} \text{ rad/s}^2 \quad \Rightarrow \quad 90^\circ \text{ rotation: } \approx \textbf{3 hours} \tag{6}$$

Scaling: $10\times$ charge or larger separation $\rightarrow$ 10-30 minute response

Orbital Maneuvering Force:

Net force from all 6 shells ($Q_{\text{total}} = 6 \times 10^{-6}$ C):

$$F_{\text{net}} = Q_{\text{total}} \cdot v \cdot B = (6 \times 10^{-6})(7670)(40 \times 10^{-6}) = 1.84 \times 10^{-6} \text{ N} \tag{7}$$

$$\Delta v: \text{ 1 m/s in 19 days; } \textbf{10-50 m/s per month} \text{ for phasing between debris} \tag{8}$$

Charging System Feasibility

Shell Design:

- Gold-plated aluminum (10 μm) with Kapton dielectric (25 μm)
- Capacitance: $C \approx 20$ pF per shell
- Energy per charge: $E = \frac{1}{2}CV^2 = \frac{1}{2}(20 \times 10^{-12})(50000)^2 = 0.025$ J
- Charging time: **Milliseconds** at 10 W power
- Leakage current: < 1 nA; Maintenance power: $P = VI = 50\ \mu$W (negligible)

Voltage Generation:

- Cockcroft-Walton cascade multiplier (5 stages)
- Converts 1 kV AC $\rightarrow$ 32-50 kV
- **Heritage:** Ion thrusters, spacecraft charging mitigation
- **TRL:** 7-8 (flight-proven technology)

Charge Control:

- *Increase charge:* Boost HVPS output (ms response)
- *Decrease charge:* Resistive discharge or electron emission

Power and Mass Budget

Component	**Mass (kg)**	**Power (W)**
6 Coulomb shells + HVPS	2.4	5-30
Electron gun system	0.5	1-10
Control electronics (CMU)	0.5	5
5 Origami sails	10.0	0
Robotic arm (optional)	15.0	20-50
Propulsion system	10.0	–
Propellant	**15.0**	–
Total (hybrid config)	**53.4**	**31-95**

Key Advantage:

- **Fuel savings: 50-70%** compared to conventional missions
- 15 kg propellant vs. 30 kg typical → enables longer missions
- Solar array: 200 W capacity sufficient (standard CubeSat tech)

Expected Results: Mission Performance

Performance Comparison:

Metric	Traditional	Magneto-Coulombic	Improvement
Debris per mission	1-2	**3-5**	**$3-5\times$**
Fuel consumption (kg)	30	**15**	**50% savings**
Mission duration (days)	30-45	**60-90**	**$2\times$**
Fragmentation risk	Medium	**Zero**	**Eliminated**
Contact required?	Yes	**Optional**	**Safer**

Projected Outcomes:

- **3-5 debris removals** per 60-90 day mission
- **Zero thruster plumes** during approach $\rightarrow$ no fragmentation
- **Months-long operations** feasible with minimal consumables
- **Lower cost-per-debris** enables viable commercial cleanup

Expected Results: Technology Validation

Anticipated Simulation Results:

1. **Lorentz Force Attitude Control**
   - 90° rotation in 10-30 minutes (with optimized charge)
   - Tumble-matching accuracy: $< 1/s$ error
   - Zero propellant consumption
2. **Orbital Phasing Capability**
   - Δv accumulation: 10-50 m/s per month
   - Gradual phasing between debris targets over days-weeks
   - Fuel reserved only for final deorbit maneuvers
3. **Electrostatic Tractor (Close Range)**
   - Force at 5m: $\sim$3.6 mN (sufficient for station-keeping)
   - 20-day towing: Cumulative $\Delta v = 50 - 100$ m/s
   - Debye shielding managed via active plasma clearing

Validation Plan:

- MATLAB/Simulink orbital dynamics simulation
- Earth's dipole magnetic field model (IGRF-13)

Limitations and Technical Challenges

1. LEO Plasma Environment (Critical Challenge)

- **Debye Shielding:** LEO plasma density $= 10^{10} - 10^{12}$ electrons/m^3
- Debye length: $\lambda_D \approx 1 - 10$ cm (vs. 1-100 m at GEO)
- **Impact:** Electrostatic forces heavily shielded beyond λ_D
- **Mitigation:**
  - Active plasma management (electron gun clears pathway)
  - Close proximity operation (5-10 m range)
  - Higher voltages (50 kV) overcome partial shielding

2. Force Magnitude Limitations

- Lorentz forces: $\sim 10^{-7}$ - 10^{-6} N (microNewton range)
- Sufficient for *gradual* attitude/orbit changes only
- **Not suitable for rapid maneuvers** (use hybrid with thrusters)

3. Magnetic Field Variations

- Earth's B-field varies with latitude, altitude, solar activity
- Requires real-time magnetometer feedback & adaptive control

Technical Risks and Mitigations

Risk	Mitigation Strategy
Insufficient torque for fast-tumbling debris ($> 30/s$)	Increase shell charge; Hybrid magneto-Coulombic + reaction wheels
Plasma arcing in LEO environment	Conformal coating; Current limiters; Redundant discharge paths
Debris material non-conductive (charging fails)	Fallback to origami sail or robotic arm options
Magnetic field unpredictability	Real-time magnetometer feedback; Adaptive control algorithms (MPC)
Long convergence times	Hybrid approach: Thrusters for initial phases, Lorentz for sustained operations

Innovation Highlights

Key Innovations:

1. **Dual-Use Lorentz Force**
   - First system enabling *both* attitude AND orbital control via charged shells
   - Eliminates need for separate reaction wheels + chemical thrusters
2. **Multi-Modal Capture Strategy**
   - Three complementary methods maximize success across debris types
   - Origami sails enable "tag-and-release" high-throughput operations
3. **Propellantless Operations**
   - 70% fuel savings $\rightarrow$ 3-5$\times$ debris removal capacity
   - Electricity-based control (exploit "free" geomagnetic momentum)
4. **Zero Fragmentation Risk**
   - No thruster plumes during close approach
   - Contactless tractor option available

Technology Spin-offs:

- Formation flying, orbit maintenance, satellite servicing
- Electrodynamic tether concepts, magnetic sailing

Impact on Space Sustainability

Addressing the Hackathon Goal: *Safe, Accessible, Usable Orbital Environment*

Scalability:

- 3-5 debris per mission
- vs. 1 for traditional systems
- Enables large-scale cleanup

Safety:

- Zero fragmentation risk
- Contactless operations
- Reduced collision hazards

Economics:

- Lower cost-per-debris
- Viable commercial cleanup
- Extended mission lifetimes

Sustainability:

- Minimal consumables
- Months-long operations
- Reusable technology platform

Vision: Transform debris removal from single-target to **fleet-scale operations**, making long-term LEO sustainability economically and technically achievable.

Conclusion

Magneto-Coulombic Actuation: A Paradigm Shift

- **Core Innovation:** Exploit Earth's magnetic field as "free" momentum source
- **Physics:** Lorentz force ($F = q(v \times B)$) enables propellantless control
- **Methodology:** Hybrid approach with 3 capture options maximizes flexibility
- **Expected Results:** 3-5 debris per mission, 50-70% fuel savings
- **Limitations:** Plasma shielding, force magnitudes, hybrid approach needed

Feasibility Confirmation:

- Sufficient force generation (μN to mN range) for attitude & orbital control
- Heritage technology (TRL 7-8 components)
- Modular design provides mission flexibility

Impact: From propellant-intensive to electricity-based space operations, enabling economically viable large-scale debris cleanup critical for long-term space sustain-ability

Thank You!

Questions?

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